Project Progress Report

Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

Michael Fang — Advisor: Prof. Salvatore Torquato, Princeton University

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1 Project Overview

This project provides a systematic numerical characterization of hyperuniform point patterns in one dimension. Two complementary metrics are computed for each pattern:

- **Hyperuniformity exponent** α: characterizes how $S(k) \sim |k|^\alpha$ as $k \to 0$.
- **Surface-area coefficient** $\bar{\Lambda}$: measures residual disorder within the strongest (Class I) patterns; $\bar{\Lambda} = 1/6$ is the global minimum (integer lattice).

Three universality classes arise from the large-R behavior of the number variance $\sigma^2(R)$:

Class	Condition	Variance	$\bar{\Lambda}$
I	$\alpha > 1$	bounded	finite
II	$\alpha = 1$	$\sim \Lambda_{II} \ln R$	diverges
III	$0 < \alpha < 1$	$\sim \Lambda_{III} R^{1-\alpha}$	diverges

The project is organized by "weeks," each corresponding to a checkpoint meeting with Prof. Torquato (not necessarily one calendar week). Three such meetings have taken place to date, yielding 27 characterized patterns and a Junior Paper (§1–§3 complete, 9 pages; results sections pending). A fourth meeting cycle is now underway.

2 Literature Review

The literature most relevant to the present project falls into four layers: (i) the definition of hyperuniformity and its order metrics, (ii) one-dimensional substitution tilings and quasicrystals, (iii) comparison families such as perturbed lattices and stealthy systems, and (iv) measurement methods based on spreadability. Read together, these papers suggest that the project is not simply "finding more examples" of hyperuniform patterns; the deeper problem is how to organize the one-dimensional landscape using both an asymptotic class parameter (α) and an intra-Class-I ranking metric ($\bar{\Lambda}$).

2.1 Foundational Framework: Hyperuniformity and Order Metrics

Torquato & Stillinger (2003). This is the foundational reference for the project. It defines hyperuniformity as $S(\mathbf{k} \to \mathbf{0}) = 0$ and proves the equivalent large-window criterion that the local

number variance grows more slowly than the window volume. For statistically homogeneous point patterns, the paper establishes the real-space and Fourier-space formulas linking $\sigma^2(R)$, $g_2(r)$, and $S(k)$. Most importantly for the present work, it identifies the **surface-area coefficient** $\bar{\Lambda}$ as the natural order metric within Class I and proves that the one-dimensional integer lattice attains the exact minimum $\bar{\Lambda}_{\min} = 1/6$. This gives the project both its baseline comparison pattern and its guiding idea that bounded-variance systems still require a secondary metric beyond hyperuniform/nonhyperuniform.

Torquato (2018). The Physics Reports review broadens the 2003 framework into a general theory of *hyperuniform states of matter*. It collects examples from crystals, quasicrystals, jammed matter, photonic materials, biological tissues, and cosmology, while emphasizing the three universality classes defined by the small-k behavior of $S(k)$ or, equivalently, the large-R behavior of $\sigma^2(R)$. For the present project, this review is important less for specific formulas than for scope: it makes clear that one-dimensional substitution tilings should be viewed as a clean laboratory for a much broader structural question, namely how long-wavelength order can coexist with strong local disorder.

Zachary & Torquato (2009). This paper extends the theory to two-phase heterogeneous media and, crucially, develops the **theta-function / Bragg-sum method** for computing $\bar{\Lambda}$ analytically from diffraction data. For this project, its most important role is practical: it provides the published Fibonacci benchmark $\bar{\Lambda} = 0.20110$ and establishes a standard route from Bragg peaks to variance-based order metrics. It also motivates the project's repeated distinction between point-pattern hyperuniformity and its two-phase embedding, a distinction that becomes essential when spreadability is used.

2.2 Substitution Tilings and Quasicrystals

Shechtman et al. (1984). The discovery of quasicrystals is not directly a one-dimensional result, but it supplies the historical reason substitution tilings matter at all. The observation of sharp diffraction with forbidden rotational symmetry established that long-range order need not imply periodicity. For the present project, that fact justifies treating one-dimensional substitution tilings as genuine models of quasicrystalline order rather than as merely combinatorial sequences.

Öguz, Macé, Torquato (2017). This paper studies hyperuniformity in projection-method quasicrystals and is particularly important for the project's interpretation of the Fibonacci chain. It shows analytically that the ideal Fibonacci strip has $Z(k) \sim k^4$ and hence $\alpha = 3$, while nonideal strips can fall to $\alpha = 1$. The paper also derives a useful upper bound $\sigma^2(R) \leq 1/4$ for Fibonacci. Conceptually, this paper matters because it ties the exponent α not just to a substitution rule, but to the underlying geometric construction of the quasicrystal. It therefore supports the project's use of Fibonacci as both a substitution chain and a cut-and-project benchmark.

Öguz, Socolar, Steinhardt, Torquato (2019). This is the central paper for the project's substitution-tiling program. Its key result is the **eigenvalue formula**

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln \lambda_1},$$

which expresses the hyperuniformity exponent entirely in terms of the substitution matrix spectrum. The paper also distinguishes Pisot and non-Pisot behavior, showing that the former is associated with Bragg diffraction and the latter with more diffuse spectral types. For this project, the paper provides the main algebraic bridge from symbolic dynamics to measurable fluctuation exponents. At the same time, it leaves open a large structural question that the current project has gradually

moved toward: once α is fixed by the matrix spectrum, what controls the *magnitude* of residual Class I disorder as measured by $\bar{\Lambda}$?

Bombieri & Taylor (1986). Bombieri and Taylor studied the cubic substitution $a \to aac$, $b \to ac$, $c \to b$, showing that it has a well-defined diffraction pattern with Bragg peaks. In the language of the later eigenvalue formula, this chain has $\lambda_1 \approx 2.247$ and $|\lambda_2| \approx 0.802$, giving $\alpha = 1.545$. Its importance for the project is therefore twofold: historically it is a classic quasicrystal diffraction example, and structurally it supplies the first natural one-dimensional occupant of the interval $1 < \alpha < 2$. That makes it the key counterpoint to the metallic-mean family, which sits rigidly at $\alpha = 3$.

2.3 Comparison Families: Perturbed Lattices and Stealthy Systems

Klatt, Kim, Torquato (2020). This paper introduces the uncorrelated random lattice (URL), in which each lattice site is displaced independently and uniformly. In one dimension it provides an exact closed form

$$\bar{\Lambda}(a) = \frac{1 + a^2}{6},$$

and shows that $\alpha = 2$ for all $a > 0$. The most relevant idea for the present project is **cloaking**: at integer a, all nonzero Bragg peaks vanish even though the underlying lattice order remains encoded in higher structure. For the project, URL plays three roles at once: (i) an exact validation family, (ii) the canonical stochastic example at $\alpha = 2$, and (iii) a reference scale for comparing deterministic chains at moderate and large n.

Batten, Stillinger, Torquato (2008). This paper introduced the collective-coordinate construction of stealthy hyperuniform ground states by directly suppressing $S(k)$ inside a prescribed reciprocal-space region. Its importance here is methodological: it shows that one can generate hyperuniformity "from Fourier space downward," in contrast to substitution tilings, which generate it "from inflation rules upward." That contrast is useful for the project because it prevents the discussion of hyperuniformity from collapsing into substitution-specific language.

Torquato, Zhang, Stillinger (2015). The Physical Review X paper develops the ensemble theory for stealthy hyperuniform disordered ground states and introduces the control parameter

$$\chi = \frac{M(K)}{d(N-1)}.$$

For $\chi < 1/2$ the ground states are disordered and highly degenerate; for $\chi > 1/2$ crystalline order dominates. In $d = 1$ the paper tabulates reference values of $\bar{\Lambda}$ for stealthy systems, making it directly useful for this project's ranking comparisons. More broadly, it shows that perfect small-k suppression does *not* determine a unique local structure, reinforcing the project's emerging view that hyperuniformity classification and order ranking are distinct problems.

Morse, Steinhardt, Torquato (2024). This recent paper sharpens that point by comparing ordered and disordered stealthy states across dimensions and by emphasizing the order metric τ as a complement to $\bar{\Lambda}$. Its lesson for the present project is conceptual rather than computational: small-k constraints alone do not determine the full degree of structural order. That perspective supports the project's shift from asking only "what is α?" to also asking "where does a pattern sit in the residual-disorder ranking?"

2.4 Measurement Papers: Spreadability and Its Limits

Torquato (2021) and Wang & Torquato (2022). These papers introduce and operationalize **diffusion spreadability** as a dynamic probe of microstructure. If $E(t) = \mathcal{S}(\infty) - \mathcal{S}(t)$ denotes the excess spreadability, then for ordinary hyperuniform media one expects

$$E(t) \sim t^{-(d+\alpha)/2},$$

while stealthy systems instead display exponential decay. For the current project, these papers justify using spreadability as a second route to α, especially when direct fitting of $S(k)$ is awkward. At the same time, they also foreshadow a limitation that the project has encountered in practice: for substitution tilings with dense Bragg spectra, finite-time oscillations can seriously contaminate slope extraction.

Hitin-Bialus et al. (2024). This paper is the strongest published demonstration to date that spreadability can recover high-precision hyperuniformity exponents for quasiperiodic tilings by carefully exploiting self-similarity. It confirms Fibonacci and Period-Doubling with essentially exact precision and extends the approach to Penrose in two dimensions. For the present project, the paper is important in two opposite ways: it validates spreadability as a serious measurement tool, but it also raises the bar for when one should trust it. The message is that spreadability can work extremely well, but only when the asymptotic window is handled with substantial care.

Yuan & Torquato (2026). The 2026 preprint revisits spreadability asymptotics and derives improved fitting forms that incorporate correction terms beyond the naive log-slope method. Its relevance to the current project is immediate because it suggests a path for rescuing chains like Bombieri–Taylor, where ordinary power-law fitting converges very slowly. At the same time, the paper explicitly notes that dense-Bragg quasicrystals are not fully covered by the simplest regular-expansion theory. This is exactly the kind of nuance the project now needs: spreadability is useful, but it should be treated as an auxiliary diagnostic rather than the sole arbiter of α.

2.5 What the Literature Leaves Open

Taken together, the literature supplies the project's core ingredients but not its full organizing structure. The foundational papers define hyperuniformity, the substitution papers explain how α emerges from matrix spectra, the URL and stealthy papers provide comparison families with exact or ensemble-based benchmarks, and the spreadability papers offer a dynamic measurement route. What remains relatively underdeveloped in the published record is the *joint* organization of one-dimensional patterns by both:

1. a **classification axis** (α, or its Class I/II/III consequences), and
2. a **ranking axis** ($\bar{\Lambda}$ within Class I).

In particular, the literature gives exact or near-exact values for a few benchmark systems (integer lattice, Fibonacci, URL, stealthy ensembles), but does not yet provide a comprehensive numerical map of how deterministic substitution chains compare to stochastic and stealthy families on the same $\bar{\Lambda}$ scale. This is the gap the present project has gradually moved to occupy.

3 Week 1

3.1 Code Benchmarking and Validation

The core numerical algorithm for computing $\sigma^2(R)$ was validated against the 1D Poisson process ($\sigma^2 = 2\rho R$) to within **1.1% mean relative error** across the full R-range. Output: `fig1_poisson_benchmark.png`.

3.2 Quasicrystal Generation

Two generation methods were implemented:

1. **Substitution method** (`substitution_tilings.py`): iterative rule application up to $N \sim 10^7$ for all five metallic-mean chains.
2. **Cut-and-project method** (`projection_method.py`): Fibonacci chain from $\mathbb{Z}^2$ projection. Agreement with substitution: $< 0.1\%$ on $\bar{\Lambda}$.

Chain	**Metallic mean** μ_n	$\lvert\det M\rvert$	α (exact)
Fibonacci ($n = 1$)	$(1+\sqrt{5})/2 \approx 1.618$	1	3
Silver ($n = 2$)	$1+\sqrt{2} \approx 2.414$	1	3
Bronze ($n = 3$)	$(3+\sqrt{13})/2 \approx 3.303$	1	3
Copper ($n = 4$)	$2+\sqrt{5} \approx 4.236$	1	3
Nickel ($n = 5$)	$(5+\sqrt{29})/2 \approx 5.193$	1	3

All 2×2 metallic-mean matrices have $\lvert\det M\rvert = 1$, which forces $\lvert\lambda_2\rvert = 1/\lambda_1$ and thus $\alpha = 3$ exactly for all of them.

3.3 Number Variance and $\bar{\Lambda}$

$\sigma^2(R)$ was computed for all five metallic-mean chains at $N \sim 10^6$–10^7. All are Class I (bounded, oscillating variance). The exact formula for the integer lattice is $\sigma^2(R) = \{R\}(1 - 2\{R\})$ where $\{R\}$ denotes the fractional part.

Pattern	$\bar{\Lambda}$	**Notes**
Integer Lattice	$1/6 = 0.1667$	Literature minimum
Fibonacci	0.2011 ± 0.001	Confirms Zachary & Torquato (2009)
Silver	0.2500	Novel (later confirmed $= 1/4$ exactly; see §4)
Bronze	0.2821 ± 0.0006	Novel
Copper	0.293	Novel
Nickel	0.310	Novel

3.4 Spreadability

Point patterns are embedded into two-phase media ($\phi_2 = 0.35$). The excess spreadability $E(t) = \mathcal{S}(\infty) - \mathcal{S}(t)$ yields α from the long-time slope $E(t) \sim t^{-(1+\alpha)/2}$.

Pattern	α (measured)	α (expected)	Error
Fibonacci	3.049	3	1.6%
Silver	2.992	3	0.3%
Bronze	2.987	3	0.4%

Integer lattice confirmed $\alpha \to \infty$ (exponential decay). Spreadability was validated as a second, independent measure of α.

4 Week 2

4.1 Uncorrelated Random Lattice and Stable Distributions

The uncorrelated random lattice (URL) displaces each site by $U(-a, a)$. The exact formula $\bar{\Lambda}(a) = (1 + a^2)/6$ was derived and validated to $< 0.3\%$ for all a. URL gives $\alpha = 2$ for all $a > 0$ (Class I). At $a = 1$ (cloaking limit), $\bar{\Lambda} = 1/3$ exactly — a useful reference value when comparing large-n metallic-mean chains.

Symmetric α-stable perturbations fill the $1 < \alpha < 2$ space stochastically (Class II/III for stability index $s < 1$).

4.2 Stealthy Hyperuniform Patterns

Stealthy configurations ($S(k) = 0$ for $|k| < K^*$) were analyzed:

- Collaborator-provided ensemble: ∼4300 disordered-phase configurations per χ ($N = 2000$, $\rho = 1$, $\chi \in \{0.1, 0.2, 0.3\}$).
- 1D approximation derived: $\bar{\Lambda} \approx 1/(\pi^2\chi)$, validated below.

χ	$\bar{\Lambda}$ (measured)	$\bar{\Lambda} = 1/(\pi^2\chi)$	Excess
0.10	1.021	1.013	+0.8%
0.20	0.526	0.507	+3.7%
0.30	0.357	0.338	+5.6%

Two-phase discovery: Generating stealthy configurations via collective-coordinate optimization from a *perturbed lattice* starting point gives $\bar{\Lambda} \approx 1/6$ — near the lattice minimum — while collaborator data (random starting positions) gives $\bar{\Lambda} \approx 1/(\pi^2\chi) \gg 1/6$. Both phases satisfy $S(k) = 0$ for $|k| < K^*$ but differ drastically in their spatial structure. This shows stealthy hyperuniformity does not uniquely determine a point pattern.

Re-classification: Stealthy patterns are not power-law suppressed; α is not meaningful. The correct descriptor is $K^* = 4\pi\rho\chi$.

4.3 New Substitution Chains

Three new chains were added to `substitution_tilings.py`:

Chain	**Rules**	**Matrix**	**Class**	α
Period-Doubling	$S \to L$, $L \to LSS$	$[[0, 2], [1, 1]]$	II	1.000
0222 Chain	$S \to LL$, $L \to SSLL$	$[[0, 2], [2, 2]]$	III	0.639
Bombieri-Taylor	$a \to aac$, $b \to ac$, $c \to b$	3×3	I	1.545

Bombieri-Taylor fills the $1 < \alpha < 2$ gap:

$$M_{BT} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_1 = 2.247, \quad |\lambda_2| = 0.802, \quad \alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = \mathbf{1.545}, \quad \bar{\Lambda} = 0.377$$

This is the first known Class I quasicrystal with $1 < \alpha < 2$.

A new $\alpha = 2$ chain was also added (first deterministic substitution tiling at this value, previously only the stochastic URL occupied $\alpha = 2$):

$$a \to bc,\ b \to c,\ c \to abc \quad \Rightarrow \quad \lambda_1 \approx 2.148,\ |\lambda_{2,3}| \approx 0.682 \text{ (complex pair)},\ \alpha = 2.000,\ \bar{\Lambda} = 0.275.$$

4.4 Full Catalog Assembly and Junior Paper

All patterns were compiled into `catalog.json` via `research_catalog.py`. Infrastructure: `disordered_patterns.p` `stealthy_analysis.py`, `make_figures.py`.

The Junior Paper (`jp/jp_hyperuniformity.tex`) was drafted: §1 Introduction, §2 Theoretical Background, §3 Methods (5 subsections). 9 pages, biblatex+biber backend. Results sections (§4–§8) exist as a separate draft (`jp/jp_results_draft.tex`) pending organization.

5 Week 3

Work completed between the second and third meetings.

5.1 Spreadability Convergence for Bombieri-Taylor

Figure: `fig_bt_deep.png`

Spreadability converges to $\alpha = 1.545$ for BT, but very slowly: $\hat{\alpha} = 2.26$ at $t \in [10^2, 10^5]$, 1.98 at $[10^4, 10^7]$, 1.51 at $[10^8, 10^{11}]$. Two causes:

1. Large $|\lambda_2| = 0.802$ (vs. 0.618 for Fibonacci) means correction terms in the spreadability decay slowly.
2. BT has 24× more Bragg peaks than Fibonacci (10^5 vs. 4,300), causing oscillations that bias slope fits.

$\bar{\Lambda}$ from $\sigma^2(R)$ avoids these issues (no fitting window needed); $\bar{\Lambda}_{\mathrm{BT}} = 0.377$.

5.2 Finite-Size Convergence

Figure: `fig_finite_size_alpha.png`

$\bar{\Lambda}$ from $\sigma^2(R)$ is stable from $N \sim 10^3$ for both BT and Fibonacci. BT spreadability $\hat{\alpha}$ converges from ∼2.4 toward 1.545 but needs $t > 10^8$. Fibonacci spreadability works at moderate t.

5.3 3×3 Substitution Matrix Search

Figure: `fig_new_tilings.png`

Enumerated all 39,304 3×3 non-negative integer matrices with $|\det M| = 1$ and row sums ≤ 4. Of these, 4,882 satisfy $|\lambda_i| < 1$ for $i \geq 2$. Among those, 504 have complex conjugate eigenvalue pairs; all 504 give $\alpha = 2$ exactly (18 distinct λ_1 values, range 1.32–3.85). The remaining 4,378 matrices (all-real eigenvalues) give $\alpha = 3$. No matrix gives $\alpha \in (2, 3)$.

The $\alpha = 2$ result follows from the determinant constraint: for 3×3 with $|\det M| = 1$ and complex pair $\lambda_2 = \bar{\lambda}_3$,

$$\lambda_1 |\lambda_2|^2 = 1 \;\Rightarrow\; |\lambda_2| = \lambda_1^{-1/2} \;\Rightarrow\; \alpha = 2 \text{ exactly.}$$

Example: $a \to bc$, $b \to c$, $c \to abc$ ($\lambda_1 \approx 2.148$, $\bar{\Lambda} = 0.274$). $\bar{\Lambda}$ varies widely across the 504 matrices (0.26 to 1.7), even among those sharing the same λ_1.

5.4 Generalized Metrics for Classes II and III

Figure: `fig_generalized_metrics.png`

$\bar{\Lambda}$ is not defined for Classes II and III ($\sigma^2(R)$ grows without bound). Instead:

$$\Lambda_{II} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R} \qquad \text{(Class II)}$$

$$\Lambda_{III} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}} \qquad \text{(Class III)}$$

Extracted via curve fit: $\sigma^2 = \Lambda_{II} \ln R + b$ (Class II) and $\sigma^2 = \Lambda_{III} R^{1-\alpha} + b$ (Class III, α fixed to theory). Results: Period-Doubling $\Lambda_{II} = 0.082$; 0222 chain $\Lambda_{III} = 0.143$ (with $\alpha = 0.639$ from eigenvalue formula).

5.5 The $2 < \alpha < 3$ Gap

Figure: `fig_nonunimodular.png`

Extended the 3×3 search to 4×4: 24M matrices tested, 1.36M with $|\lambda_i| < 1$, zero with $\alpha \in (2, 3)$. The gap persists because:

- Rank 3 with complex pair: $|\det M| = 1$ forces $\alpha = 2$ (proof above)
- Rank ≥ 4 with complex pair: extra small eigenvalues force $|\lambda_2|$ larger, giving $\alpha < 2$
- All-real eigenvalues: numerical search finds only $\alpha = 3$

No substitution matrix with $|\det M| = 1$ at any rank can produce $\alpha \in (2, 3)$.

Non-unit determinant: For 2×2 matrices with $|\lambda_2| < 1$:

$$\alpha = 3 - \frac{2 \ln |\det M|}{\ln \lambda_1}$$

To get $\alpha \in (2, 3)$: need $1 < |\det M| < \sqrt{\lambda_1}$. Since det is an integer, only $|\det M| = 2$ works (requires $\lambda_1 > 4$). Found 28 such matrices, all with $|\det M| = 2$ and $\alpha \in (2.0, 2.2)$.

Smallest example: $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$), rules $a \to b$, $b \to aabbbb$. $\lambda_1 = 2 + \sqrt{6} \approx 4.449$, $\alpha = 2.071$. $\sigma^2(R)$ oscillates over $[0.001, 0.87]$ (Fibonacci: $[0.15, 0.25]$). $\bar{\Lambda} \approx 0.648 \pm 0.004$ ($N = 17.8$M, period-averaged). First known 1D Class I chain with $\alpha \in (2, 3)$.

5.6 Silver $\bar{\Lambda} = 1/4$ and Metallic-Mean Convergence

Figures: `fig_silver_lambda_quarter.png`, `fig_metallic_convergence.png`

Silver: $N = 9.37 \times 10^6$ points, 5×10^4 windows: $\bar{\Lambda} = 0.2502 \pm 0.0006$ (0.3σ from 1/4). Silver has $\rho = 1/2$ exactly. Fibonacci: 0.201 (ours) vs. 0.20110 (Zachary & Torquato 2009, theta-series; both numerical). Bronze: 0.2822 ± 0.0006 (no obvious closed form).

Metallic-mean sequence: Computed $\bar{\Lambda}(\mu_n)$ for $n = 1$ through 20:

n	1	2	3	4	5	6	7	8	10
$\bar{\Lambda}$	0.201	0.250	0.282	0.293	0.310	0.318	0.321	0.325	0.327
n	15	20	URL a=1						
$\bar{\Lambda}$	0.329	0.331	1/3						

For $n \leq 20$, the measured values increase monotonically and approach the cloaked-URL benchmark $\bar{\Lambda} = 1/3$ from below. As $n \to \infty$, $L/S = \mu_n \to \infty$ but $\ell_L/\langle \ell \rangle \to 1$: the long tile becomes the mean spacing, short tiles become rare and tiny, and the chain approaches a lattice with disorder matching URL $a = 1$.

6 Week 4

Tasks from the third meeting (2026-03-27) with Prof. Torquato.

6.1 Why $\alpha_{\max} = 3$ for 1D Substitution Tilings

The eigenvalue formula $\alpha = 1 - 2\ln|\lambda_2|/\ln\lambda_1$ (Öguz et al. 2019) gives $\alpha = 3$ for all metallic-mean chains. The bound $\alpha \leq 3$ holds for *all* unimodular ($|\det M| = \pm 1$) substitution matrices, for the following reason.

For a rank-r substitution matrix M with Perron–Frobenius eigenvalue λ_1 and remaining eigenvalues $\lambda_2, \ldots, \lambda_r$, the determinant constraint gives

$$|\det M| = \lambda_1 \prod_{i=2}^{r} |\lambda_i| \geq \lambda_1 |\lambda_2|^{r-1} \quad \Rightarrow \quad |\lambda_2| \leq \left(\frac{|\det M|}{\lambda_1}\right)^{1/(r-1)}.$$

Substituting into the eigenvalue formula:

$$\alpha = 1 - \frac{2\ln|\lambda_2|}{\ln\lambda_1} \leq 1 + \frac{2}{r-1}\left(\frac{\ln\lambda_1 - \ln|\det M|}{\ln\lambda_1}\right) = 1 + \frac{2}{r-1}\left(1 - \frac{\ln|\det M|}{\ln\lambda_1}\right).$$

For $|\det M| = 1$ this simplifies to $\alpha \leq 1 + 2/(r-1)$. The bound is saturated when all sub-dominant eigenvalues have the same absolute value, i.e., $|\lambda_i| = \lambda_1^{-1/(r-1)}$ for $i \geq 2$.

Rank r	$\alpha_{\max}$	**Example**
2	3	all metallic means (Fibonacci, Silver, ...)
3	2	504 cubic matrices with complex pair
4	$5/3 \approx 1.67$	(no known examples saturating this)
$r \to \infty$	$\to 1^+$	approaches Class II boundary

Thus $\alpha = 3$ is the global maximum for unimodular substitution tilings in 1D, achieved uniquely by rank-2 (two-letter) chains. Higher-rank matrices necessarily have $\alpha < 3$ because the extra eigenvalues in $|\det M| = 1$ force $|\lambda_2|$ to be larger (closer to 1), reducing α. Non-unimodular matrices can exceed $\alpha = 3$ in principle (e.g. $|\det M| = 0$ gives $\alpha = \infty$, but this degenerates to a lattice), though in practice the range $\alpha \in (2, 3)$ requires $1 < |\det M| < \sqrt{\lambda_1}$ as shown in §3.5.

6.2 Physical Interpretation of the $\bar{\Lambda}$ Ranking

$\bar{\Lambda}$ quantifies residual large-scale density fluctuations: larger $\bar{\Lambda}$ means more disorder. The catalog ranking (§5) is controlled by a few structural parameters:

1. **Perfect order → minimum.** The integer lattice achieves $\bar{\Lambda} = 1/6$, the global minimum for all 1D hyperuniform patterns. Any departure from perfect periodicity increases $\bar{\Lambda}$.
2. **Metallic means: tile disparity controls disorder.** As n grows, the tile-length ratio $\mu_n = L/S$ increases. Short tiles become rarer (frequency $\sim 1/\mu_n \to 0$) and act as "defects" in an otherwise lattice-like array of long tiles. More extreme defects → larger variance → larger $\bar{\Lambda}$. For the low- and moderate-n chains studied, the sequence $\bar{\Lambda}(\mu_1) = 0.201 < \bar{\Lambda}(\mu_2) = 0.250 < \cdots$ increases overall, but its large-n behavior remains unresolved.
3. **URL: displacement amplitude a controls disorder directly.** Each site is displaced by $U(-a/2, a/2)$, giving $\bar{\Lambda} = (1 + a^2)/6$. At $a = 0$: lattice ($\bar{\Lambda} = 1/6$). At $a = 1$: cloaked ($\bar{\Lambda} = 1/3$), comparable to the large-n metallic-mean values. Intermediate a interpolates smoothly.

4. **Stealthy: weaker constraint → more disorder.** The stealth condition $S(k) = 0$ for $|k| < K^*$ constrains fewer degrees of freedom when χ is small. $\bar{\Lambda} \approx 1/(\pi^2\chi)$: smaller χ allows larger fluctuations.
5. **Bombieri-Taylor: slow decay of $S(k)$ means large fluctuations.** $\alpha = 1.545$ is low (close to Class II boundary), so $S(k)$ decays slowly near $k = 0$, allowing more long-wavelength fluctuations. Combined with $|\lambda_2| = 0.802$ (close to 1), giving $\bar{\Lambda} = 0.377$.

In summary, $\bar{\Lambda}$ increases with: (i) larger tile-length disparity in substitution chains, (ii) larger displacement amplitude in perturbed lattices, (iii) weaker spectral constraints in stealthy patterns, and (iv) slower power-law suppression of $S(k)$ near the origin.

6.3 Uncloaked URL Patterns

Figure: `fig_url_sweep.png`

The exact formula $\bar{\Lambda}(a) = (1+a^2)/6$ (Klatt et al. 2020) was validated by simulation at 17 values of $a \in [0.05, 2.0]$ with $N = 50{,}000$ points each. All simulated values match the formula to within 1%.

a	$\bar{\Lambda}$ (exact)	$\bar{\Lambda}$ (sim)	Error	Note
0	0.1667	—	—	lattice
0.5	0.2083	0.2082	0.08%	
1.0	0.3333	0.3330	0.10%	cloaked (Bragg peaks vanish)
1.5	0.5046	0.5047	0.02%	
2.0	0.6667	0.6657	0.14%	second cloaking event

Key observations: (i) $\bar{\Lambda}$ grows monotonically with a (more displacement = more disorder). (ii) At each integer $a = 1, 2, \ldots$, cloaking occurs: the Bragg peaks of the underlying lattice vanish because the displacement form factor $\tilde{f}(k) = \sin(\pi k a)/(\pi k a)$ has zeros at all non-zero reciprocal lattice vectors. (iii) $\bar{\Lambda}(a = 1) = 1/3$ provides a useful benchmark for comparison with the large-n metallic-mean data.

6.4 g_2-Invariant Patterns

Torquato & Stillinger (2003, §V.B) define a **g_2-invariant process** as one where the pair correlation function $g_2(r)$ remains invariant as density increases, up to a terminal density ρ_c at which the system becomes hyperuniform ($S(0) = 0$).

Three 1D examples were investigated:

1. **Lattice-gas:** Tessellate $[0, L)$ into N intervals of length $D = 1$; place one point uniformly at random in each interval. $g_2(r) = r/D$ for $r \leq D$, $g_2 = 1$ for $r > D$. This is equivalent to URL with $a = 1$ (just a phase offset: $j + U(0, 1)$ vs. $j + U(-1/2, 1/2)$). $\bar{\Lambda} = 0.333 = 1/3$, confirming the URL result.
2. **Step-function g_2:** $g_2(r) = \Theta(r - D)$ at terminal density $\phi_c = 1/2^d$. For $d = 1$: $\phi_c = 1/2$, close-packed hard rods = lattice. $\bar{\Lambda} = 1/6$.
3. **Gaussian shuffled lattice:** Each lattice site displaced by $\mathcal{N}(0, \sigma^2)$. Hyperuniform for any $\sigma > 0$. Measured: $\bar{\Lambda} = 0.200$ ($\sigma = 0.1$), 0.250 ($\sigma = 0.2$), 0.342 ($\sigma = 0.3$). At $\sigma = 0.1$, $\bar{\Lambda}$ matches Fibonacci; at $\sigma = 0.2$, it matches Silver = 1/4.

The g_2-invariant processes at terminal density are all hyperuniform and fall naturally into the existing ranking: they are particular cases of perturbed lattices, with $\bar{\Lambda}$ determined by the displacement

distribution. The lattice-gas is a special case of URL. The step+delta g_2 examples (Eq. 121–125 in the 2003 paper) are 3D constructions with no direct 1D analog beyond the cases already computed.

6.5 Metallic Chain Tile Visualizations

Figure: `fig_metallic_tiles_extended.png`

Figure shows 25 tiles of each metallic-mean chain for $n = 1, 2, 3, 4, 5, 6, 8, 10$. As n increases, short tiles (S) become rarer: the S fraction decreases from 36% at $n = 1$ (Fibonacci) to 8% at $n = 10$. For large n, the chain approaches a near-lattice of long tiles with rare, widely-spaced short-tile defects.

6.6 Metallic-Mean Convergence to $n = 100$

Extended $\bar{\Lambda}(\mu_n)$ from $n = 20$ to $n = 100$ (TARGET_N = 300k).

n	1	2	3	5	10	20	30	50	70	100	URL a=1
$\bar{\Lambda}$	0.199	0.250	0.283	0.310	0.327	0.332	0.332	0.334	0.330	0.335	1/3

The values for moderate n rise toward the URL benchmark 1/3, but the largest-n runs do not support treating this as a settled asymptotic limit. Some nominal overshoots at $n \geq 50$ are within the $\sim$0.005 error bars (reduced accuracy at large n because chains grow very slowly: $\lambda_1 \approx n + 1/n$, so even at $n = 100$ only two substitution iterations are needed to reach $N = 300$k, limiting the quasi-periodic structure). The overall picture is that 1/3 remains a plausible comparison value, but the present data do not establish monotone convergence to the cloaked-URL limit.

6.7 All Non-Unimodular Chains

Script: `compute_all_nonunimodular.py`

Enumerated all 2×2 non-negative integer matrices with $|\det M| = 2$, $|\lambda_2| < 1$, $\alpha \in (2, 3)$, and row sums ≤ 10. Found **138 matrices** (more than the original 28 found with row sums ≤ 6) across 12 distinct α values.

α	Count	$\bar{\Lambda}$ range	Smallest $\bar{\Lambda}$
2.071	12	$[0.32, 2.01]$	$M = \left[\begin{smallmatrix} 0 & 2 \\ 1 & 9 \end{smallmatrix}\right]$
2.087	10	$[0.34, 1.57]$	
2.175	18	$[0.33, 3.36]$	
2.199	14	$[0.34, 3.57]$	
2.248	10	$[0.33, 2.82]$	
2.271	16	$[0.34, 3.76]$	
2.301	18	$[0.33, 4.05]$	
2.343	12	$[0.33, 4.79]$	
2.362	12	$[0.34, 4.80]$	
2.376	8	$[0.33, 4.97]$	
2.393	4	$[0.34, 5.08]$	

Key observations: (i) All α values lie in $[2.07, 2.39]$, consistent with the formula $\alpha = 3 - 2\ln 2/\ln \lambda_1$ and the constraint $\lambda_1 > 4$. (ii) $\bar{\Lambda}$ varies enormously *within* each α group (by factors of 5–15), showing that α alone does not determine $\bar{\Lambda}$. (iii) The smallest $\bar{\Lambda} \approx 0.32$–$0.34$ across all groups, close to 1/3 (the URL cloaked value). (iv) These are all Class I ($|\lambda_2| < 1$, bounded σ^2).

6.8 Yuan-Torquato Improved Spreadability

Script: `yuan_improved_spreadability.py`

Implemented the improved fitting procedure from Yuan & Torquato (2026, arXiv:2602.17873). Three fitting function types were tested on Fibonacci ($\alpha = 3$):

Chain	**Method**	$\hat{\alpha}$	$\lvert\hat{\alpha} - \alpha\rvert$	Error
Fibonacci ($\alpha = 3$)	Simple log-slope	2.19	0.81	27%
	Type I, order 1	3.04	0.04	1.3%
	Type III, order 1	2.95	0.05	1.5%
BT ($\alpha = 1.545$)	Simple log-slope	2.27	0.72	47%
	Type I, order 6	1.57	0.03	**1.7%**
	Type III, order 6	2.16	0.61	40%

For Fibonacci, Type I at low order ($n = 1$) gives the best result (1.3% error); higher orders diverge because quasicrystal $s^{ex}(t)$ oscillates (dense Bragg peaks). For BT, Type I at high order ($n = 5$–6) dramatically reduces the error from 47% to $\sim 2\%$ — the **first time BT's α has been measured accurately from spreadability without extreme time ranges** ($t > 10^8$ was previously needed). Type III fails for BT because the spectral density is not analytic at the origin.

Conclusion: The Yuan-Torquato Type I fitting is a significant improvement for chains with large $|\lambda_2|$ (like BT), where correction terms dominate at moderate t. It does not replace variance-based $\bar{\Lambda}$ extraction but provides a viable alternative for α measurement from spreadability data.

7 Complete Pattern Catalog

Patterns sorted by $\bar{\Lambda}$ within Class I.

Pattern	**Class**	$\boldsymbol{\alpha}$	$\bar{\boldsymbol{\Lambda}}$
Integer Lattice	I	∞	$1/6 = 0.167$
URL ($a = 0.1$)	I	2.0	0.168
URL ($a = 0.3$)	I	2.0	0.181
Gaussian $\sigma = 0.1$	I	2.0	0.187
Fibonacci	I	3.0	0.201
URL ($a = 0.5$)	I	2.0	0.208
Silver	I	3.0	$\mathbf{1/4} = 0.250$
Cubic α=2 (new)	I	2.0	0.275
Bronze	I	3.0	0.282
Copper	I	3.0	0.293
Nickel	I	3.0	0.310
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	—†	0.358
Bombieri-Taylor	I	1.545	0.377
Stable $s = 1.7$	I	1.7	0.242
Stable $s = 1.5$	I	1.5	0.303
Stable $s = 1.3$	I	1.3	0.417
Stealthy ($\chi = 0.2$)	I	—†	0.526
Stealthy ($\chi = 0.1$)	I	—†	1.021
Period-Doubling	II	1.0	div. ($\Lambda_{II} = 0.080$)
0222 Chain	III	0.639	div. ($\Lambda_{III} = 0.209$)
Cauchy $\gamma = 0.1$	II	0.95	div.
Stable $s = 0.7$	III	0.7	div.
Stable $s = 0.5$	III	0.5	div.
Stable $s = 0.3$	III	0.3	div.

† Stealthy: $S(k) = 0$ by construction; α not applicable. Green rows: literature-confirmed benchmarks. Yellow: novel additions.

8 Figures and Scripts

Figures

Filename	Content
`fig1_poisson_benchmark.png`	Poisson variance validation
`fig2_bounded_variance.png`	$\sigma^2(R)$ for Fibonacci/Silver/Bronze
`fig3_hyperuniformity_test.png`	Normalized variance across classes
`fig9_stealthy_sk.png`	$S(k)$ for stealthy at $\chi = 0.1, 0.2, 0.3$
`stealthy_Sk_overlay.png`	$S(k)$ overlay of all stealthy configs
`bombieri_taylor_analysis.png`	BT variance, $S(k)$, spreadability
`fig12_ranking.png`	Full $\bar{\Lambda}$ ranking, Class I
`fig_variance_II_III.png`	$\sigma^2(R)$ for Period-Doubling and 0222
`fig_bt_deep.png`	BT oscillations, Bragg peak density, $E(t)$
`fig_finite_size_alpha.png`	$\bar{\Lambda}(N)$, $\Lambda_{II}(N)$, $\hat{\alpha}(N)$ (portrait)
`fig_finite_size_alpha_landscape.png`	Same, landscape 1×4 for slides
`fig_new_tilings.png`	Cubic α=2 chain variance and $S(k)$
`fig_generalized_metrics.png`	Normalized variance plateaus (Class II/III)
`fig_finite_size_lambda.png`	Lattice $\bar{\Lambda}$ analytic vs. simulated
`fig_catalog_updated.png`	Updated $\bar{\Lambda}$ ranking with stealthy fixes
`fig_metallic_convergence.png`	$\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$, compared with the URL a=1 value 1/3
`fig_4x4_alpha_dist.png`	4×4 matrix search: α distribution
`fig_nonunimodular.png`	$\sigma^2(R)$ and running $\bar{\Lambda}$ for $\alpha = 2.071$ chain

Scripts

Script	Purpose
`substitution_tilings.py`	All chains (metallic means, BT, cubic α=2, non-unimodular)
`quasicrystal_variance.py`	$\sigma^2(R)$, $\bar{\Lambda}$, generalized metrics
`projection_method.py`	Cut-and-project Fibonacci chain
`two_phase_media.py`	Spectral density, spreadability
`disordered_patterns.py`	URL, stealthy generation, exact $\bar{\Lambda}$ for URL
`stealthy_analysis.py`	Collaborator stealthy ensemble analysis
`research_catalog.py`	Full pipeline $\rightarrow$ `catalog.json`
`make_figures.py`	Figures from catalog + stealthy results
`bt_deep_dive.py`	(Mtg 2–3) BT oscillations, $S(k)$ density, $E(t)$
`finite_size_scaling.py`	(Mtg 2–3) $\bar{\Lambda}(N)$, $\Lambda_{II}(N)$, $\hat{\alpha}(N)$
`search_cubic_tilings.py`	(Mtg 2–3) Exhaustive 3×3 matrix search
`generalized_metrics.py`	(Mtg 2–3) Λ_{II}, Λ_{III}; patches variance module
`finite_size_lambda_bar.py`	(Mtg 2–3) Lattice formula; stealthy two phases
`metallic_mean_convergence.py`	(Mtg 2–3) $\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$
`search_4x4_tilings.py`	(Mtg 2–3) 4×4 matrix search (24M matrices)
`silver_precision_analysis.py`	(Mtg 2–3) High-precision Silver/Bronze $\bar{\Lambda}$
`fig_nonunimodular_chain.py`	(Mtg 2–3) Figure for $\alpha = 2.071$ non-unimodular chain

Presentations and Written Work

Document	Location	Notes
Week 1 presentation	`results/week1/week1_presentation.tex`	Beamer, core metallic-mean results
Week 2 presentation	`results/week2/week2_presentation.tex`	Beamer, 12 slides
Week 3 presentation	`results/week3/week3_presentation.tex`	Beamer, 16 slides (rewritten 2026-03-24)
Junior Paper	`jp/jp_hyperuniformity.tex`	9 pages, §1–§3 complete
JP results draft	`jp/jp_results_draft.tex`	§4–§8 + appendix, held for later

All documents compile via MiKTeX (`C:/Users/minec/AppData/Local/Programs/MiKTeX/miktex/bin/x64/`).

9 Key Theoretical Results

1. $\alpha = 3$ **universality of metallic means.** For any 2×2 metallic-mean matrix with $|\det M| = 1$: $|\lambda_2| = 1/\lambda_1 \Rightarrow \alpha = 3$ universally.

2. **Bombieri-Taylor fills $1 < \alpha < 2$.** The 3-letter cubic substitution $a \to aac, b \to ac, c \to b$ gives $\alpha = 1.545$, the first known 1D quasicrystal with $\alpha \in (1, 2)$.

3. **First deterministic $\alpha = 2$ substitution tiling.** The chain $c \to abc, a \to bc, b \to c$ gives $\alpha = 2.000$ exactly, $\bar{\Lambda} = 0.275$.

4. **$\bar{\Lambda}$ density invariance.** $\bar{\Lambda}$ is invariant under uniform density rescaling in $d = 1$; direct comparison across patterns with different densities is valid.

5. **Integer lattice has no finite-N correction.** $\bar{\Lambda} = 1/6$ exactly for all integer $R_{\max}$ (piecewise cubic integral identity).

6. **Stealthy two-phase structure.** Both ordered and disordered stealthy phases satisfy $S(k) = 0$ for $|k| < K^*$, but differ drastically in $\bar{\Lambda}$ ($\approx 1/6$ vs. $1/(\pi^2\chi)$). Stealthy α is not a power-law exponent.

7. **$2 < \alpha < 3$ gap is inaccessible to unimodular matrices.** Exhaustive search over 3×3 (39,304) and 4×4 (24M) matrices confirms zero unimodular cases with $\alpha \in (2, 3)$. Rank-3 proof: $|\det M| = 1 \Rightarrow \alpha = 2$ exactly for complex pairs. Rank-≥ 4 proof: additional Pisot factor forces $\alpha < 2$.

8. **General 2×2 formula.** For any 2×2 Pisot matrix: $\alpha = 3 - 2\ln|\det M|/\ln\lambda_1$. Gap $(2, 3)$ is accessible when $|\det M| \in (1, \sqrt{\lambda_1})$; all 28 known examples have $|\det M| = 2$.

9. **First 1D quasicrystal with $\alpha \in (2, 3)$.** $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$) gives $\alpha = 2.071$, $\bar{\Lambda} \approx 0.650$ (rough estimate). Large log-periodic oscillations in $\sigma^2(R)$, period $\ln\lambda_1 \approx 1.493$.

10. **Silver $\bar{\Lambda} = 1/4$ exact (numerical).** $N = 3.88 \times 10^6$, 3×10^4 windows: $\bar{\Lambda} = 0.2501 \pm 0.0004$ (0.3σ from $1/4$). Analytical proof unknown.

11. **Large-n metallic-mean comparison with URL.** For $n = 1 \ldots 20$, the metallic-mean sequence rises toward the cloaked-URL value $\bar{\Lambda} = 1/3$, but the asymptotic limit is not established.

10 Open Questions

1. $2 < \alpha < 3$ **gap:** ***RESOLVED*** **(mechanism and first example).** Non-unimodular $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ gives $\alpha = 2.071$. Sub-question: what is the minimal-complexity chain in $(2, 3)$? (Row sum ≥ 6 is currently the minimum requirement.)

2. **Silver exact value:** ***RESOLVED numerically.*** $\bar{\Lambda} = 1/4$ at 0.3σ. Analytical proof not yet known.

3. $\bar{\Lambda}$ **metallic-mean large-n behavior:** The data suggest a relationship to the cloaked URL value $1/3$, but the asymptotic limit remains unresolved.

4. **Non-unimodular $\bar{\Lambda}$:** Is $\bar{\Lambda} = 0.650$ exactly $13/20$? Analytical proof unknown; large oscillation amplitude makes high-precision measurement difficult.

5. **Bronze closed form:** Is $\bar{\Lambda}(\text{Bronze}) = 0.2822$ a clean expression? Closest simple fraction: $11/39 = 0.28205$ (0.18σ, non-obvious).

6. **Stealthy two-phase literature:** Is the ordered/disordered bifurcation discussed in existing stealthy ground-state literature?

7. **2D generalization:** Does the eigenvalue formula $\alpha = 1 - 2\ln|\lambda_2|/\ln\lambda_1$ extend to 2D substitution tilings (Penrose, Ammann-Beenker)?